Informatica® Cloud Data Quality
July 2026

Data Quality Dictionaries

Informatica Cloud Data Quality Data Quality Dictionaries
July 2026

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Preface

Use *Data Quality Dictionaries* to learn how to create and manage dictionaries for data quality assets. A dictionary is a reference data object that you can use in a data quality asset to verify and enhance the accuracy and usability of your data.

Informatica Resources

Informatica provides you with a range of product resources through the Informatica Network and other online portals. Use the resources to get the most from your Informatica products and solutions and to learn from other Informatica users and subject matter experts.

Informatica Documentation

Use the Informatica Documentation Portal to explore an extensive library of documentation for current and recent product releases. To explore the Documentation Portal, visit https://docs.informatica.com.

If you have questions, comments, or ideas about the product documentation, contact the Informatica Documentation team at infa_documentation@informatica.com.

IDMC web site

You can access the Informatica® Intelligent Data Management Cloud™ (IDMC) web site at http://www.informatica.com/cloud.

IDMC Communities

Use the IDMC Community to discuss and resolve technical issues. You can also find technical tips, documentation updates, and answers to frequently asked questions.

Access the IDMC Community at:

https://network.informatica.com/community/informatica-network/products/cloud-integration

Developers can learn more and share tips at the Cloud Developer community:

https://network.informatica.com/community/informatica-network/products/cloud-integration/cloud-developers

IDMC Marketplace

Visit the Informatica Marketplace to try and buy Data Integration Connectors, templates, and mapplets:

https://marketplace.informatica.com/

Informatica Knowledge Base

Use the Informatica Knowledge Base to find product resources such as how-to articles, best practices, video tutorials, and answers to frequently asked questions.

To search the Knowledge Base, visit https://search.informatica.com. If you have questions, comments, or ideas about the Knowledge Base, contact the Informatica Knowledge Base team at KB_Feedback@informatica.com.

IDMC Trust Center

The IDMC Trust Center provides information about Informatica security, compliance, and privacy practices.

You can access the trust center at https://trust.informatica.com/.

IDMC Pulse Dashboard

The IDMC Pulse Dashboard provides system status and upgrade information for all PODs and services across IDMC.

Follow the services you use and subscribe to alerts to receive notifications about upgrades, maintenance, production incidents, and upcoming releases. The IDMC Pulse Dashboard displays the production status of all the IDMC products. All maintenance updates are posted to this page, and during an outage, it will have the most current information. To ensure you are notified of updates and outages, you can subscribe to receive updates for a single service or all IDMC services. Subscribing to all services is the best way to be certain you never miss an update.

For more information, see the Service Status and Communications guide.

Informatica Global Customer Support

You can contact a Global Support Center through the Informatica Network or by telephone.

To find online support resources on the Informatica Network, click **Contact Support** in the IDMC Help menu to go to the **Cloud Support** page. The **Cloud Support** page includes system status information and community discussions. Log in to Informatica Network and click **Need Help** to find additional resources and to contact Informatica Global Customer Support through email.

The telephone numbers for Informatica Global Customer Support are available from the Informatica web site at https://www.informatica.com/services-and-training/support-services/contact-us.html.

CHAPTER 1

Introduction to dictionaries

A dictionary is a reference data set that you can use to evaluate data in a mapping. Use dictionaries to verify that the data values on a data source or another object in a mapping are accurate and correctly formatted.

A dictionary can contain two or more columns. At least one column must contain the standard or preferred versions of a set of values that the organization uses. Other columns can contain alternative or related versions of the values, including versions that might appear in the source data. The column that contains the standard or preferred values is called the valid column. You can populate a dictionary with any combination of values that suits your organization and your project requirements.

To use a dictionary, add it to a data quality asset. Then, add the asset to the corresponding transformation in a mapping in Data Integration.

When you run a mapping that contains the transformation, the mapping compares each input data value to the dictionary data and performs actions that you specify. For example, a rule specification might return a value that indicates a match between the input data value and the dictionary data. Or, a cleanse asset might return a preferred or correct version of the input value.

Types of dictionary

You can create a dictionary through the Data Quality, Data Integration, or Reference 360 user interface or through the Data Quality Dictionary API. In Data Integration or Data Quality, you configure the columns and manually add values to the dictionary. In Reference 360, you can use code value attributes or value mappings as columns and their corresponding values as column values. You can view the dictionaries that you create on the **Explore** page.

Regardless of whether you created the dictionary through the Data Quality, Data Integration, or Reference 360 user interface or through the Data Quality Dictionary API, it will function the same way when added to a data quality asset.

If you create a dictionary in Reference 360, you can view the contents of the dictionary in Data Quality or Data Integration, but to edit the dictionary data, you must open the dictionary source in Reference 360.

For more information about how to create a dictionary in Reference 360, see the *Reference 360* help in Reference 360.

For more information about how to create a dictionary through the Data Quality Dictionary API, see the *API Reference* help in Data Quality.

Using a dictionary

You can create dictionaries with any data that is useful to the data project that you work on. The dictionary does not evaluate the data that you enter, and the values in each column do not need to be objectively correct or incorrect.

Consider the following scenarios:

- The dictionary might include alternative versions of the same data, such as *Massachusetts* and *MA*.
- The dictionary might include values that relates to each other in a way that is relevant to your business, such as *Massachusetts* and *Boston*.
- The dictionary might include correct and incorrect spellings of the same data, such as *Massachusetts* and *Massachusets*.

Each row in a dictionary can have a different number of values. For example, the number of telephone area codes in each state of the United States varies from state to state. If you create a dictionary that matches the area codes to their respective states, the rows for each state will have different numbers of values.

The following table shows a fragment of a dictionary of United States telephone area codes:

Column 1	Column 2	Column 3	Column 4	Column 5	Column 6	Column 7
Mississippi	228	601	662	769		
Missouri	314	417	573	636	660	816
Montana	406					
Nebraska	308	402	531			
Nevada	702	725	775			
New Hampshire	603					

Note: Column 1 is the valid column in the table of dictionary data. You might configure an asset to return the value from column 1 when it finds a value from one of the other columns in the input data.

Dictionary operations

When you run a mapping with a transformation that specifies a dictionary, the transformation compares the input field data to the data in the dictionary. If the transformation finds a match between an input value and a dictionary value, the transformation performs an action that the asset in the transformation defines. For example, the transformation might return the value in the valid column that corresponds to the input value.

Valid column behavior

You can select any column with values that suit your data objectives as the valid column. You might specify different columns as valid in different projects. The contents of the dictionary columns determine if the dictionary can support different valid columns in different business cases.

A transformation searches every column in the dictionary except the valid column that the asset specifies. If you want the search operation to include the valid column data, add a copy of the valid column to the dictionary. You can create a dictionary that contains two identical columns of data.

Note: A test that you run in a data quality asset reads a dictionary in the same way as a transformation reads a dictionary.

You can design a useful dictionary with multiple occurrences of a single value in the valid column. For example, you might design a dictionary that identifies a list of cities by time zone. Avoid designing a dictionary with multiple occurrences of a single value in non-valid columns.

Dictionary download considerations

The Secure Agent can download a dictionary that an asset specifies when you test the asset or run a mapping or profile that reads the asset.

The Secure Agent uses the following guidelines to download dictionary data:

- When you test a step in a cleanse asset that reads a dictionary, the Secure Agent downloads every dictionary that the cleanse asset specifies.

  When you test any other data quality asset, the Secure Agent downloads any dictionary that the test logic requires. If the asset specifies a dictionary that is not required during the test, the Secure Agent does not download the dictionary.
- When you run a profile or a mapping than references an asset that reads a dictionary, the Secure Agent downloads every dictionary that the asset specifies.

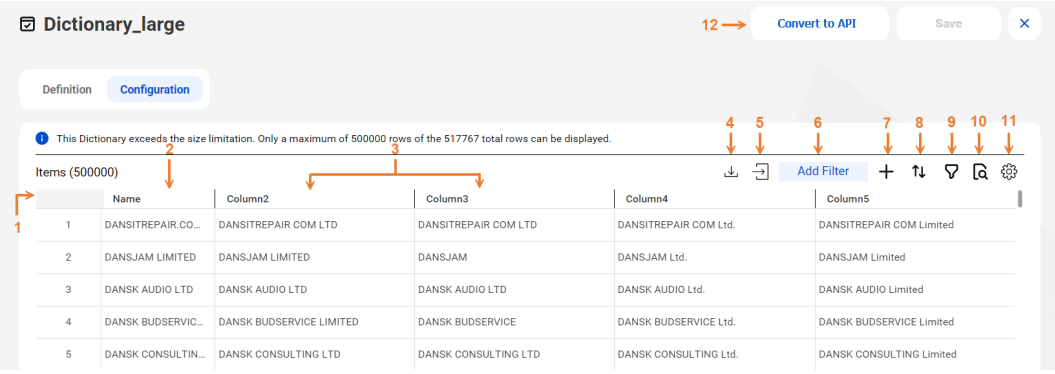
The Secure Agent downloads dictionaries to a directory on its host machine. You may decide to periodically check the download directory, for example to review the amount of disk space that the directory uses. For more information about dictionary downloads, see "Verifying the dictionary download location" on page 16.

Dictionary configuration options

On the **Configuration** view, define and update the content and structure of a dictionary.

You can select the **Configuration** view when you create or open a dictionary.

The following image shows the options that you can use to configure the dictionary:



The dictionary contains the following options and elements:

1. Row ID.
   Shows the current sequence of the dictionary rows. To highlight a row, select its row ID.

2. Default valid value column.
   Contains the values that the transformation returns at run time by default. You can specify a different valid column when you configure the asset.
3. Alternative value columns.
   Contains alternative or incorrect versions of the values in the default valid column.
4. Download data to a file.
   Downloads the data in a dictionary to a comma-separated file.
5. Import data from a file.
   Imports data to the dictionary.
6. Add a filter query.
   Opens the **Add Filter** query options.

   The **Add Filter** query option appears in a dictionary that contains very large quantities of data. For information about very large dictionaries, see "Working with very large dictionaries" on page 14.
7. Add a row.
   Adds an empty row after the current final row in the dictionary.
8. Sort the dictionary data.
   Sorts the data rows by values in one or more columns that you specify.
9. Filter the current page of dictionary data.
   Displays the rows that contain a value that you specify in a column that you select. The option applies a filter on the current page in a dictionary.
10. Find and replace data.
    Finds and replaces a value that you specify.

    Note: The Filter and Find and Replace options are linked. When the filter is active, you can find and replace data within the filtered results only.
11. Settings menu.
    Determines the dictionary row height.
12. Conversion options.
    Enables the dictionary for updates through the Data Integration or Data Quality user interface or updates through the Dictionary API. The option displays **Convert to UI** or **Convert to API** based on the current status of the dictionary. For more information about the dictionary conversion options, see "Dictionary conversion options" on page 12.

Note: If you create a dictionary in Reference 360, you can view the contents of the dictionary in Data Quality or Data Integration, but to edit the dictionary data, you must open the dictionary source in Reference 360.

Creating a dictionary

Create a dictionary by using reference data and configure the dictionary to an asset.

The **Dictionary** window contains a **Definition** tab and a **Configuration** tab. Use the **Definition** tab to define the name and the location of the dictionary. Use the **Configuration** tab to build the dictionary structure and add data.

Note: You can also open the **Definition** and **Configuration** tabs when you open the asset from the **Explore** page.

1. On the **Definition** tab, enter a name for the dictionary.

2. Optionally, enter a description.
3. Select a location for the dictionary.

   You can ignore the **Asset References** fields when you create a dictionary. A new object contains no asset references.
4. Save the dictionary.

   The **Asset References** fields are replaced with fields that include the creation date and the name of the asset creator.
5. Optionally, add a tag to the dictionary. You can search for assets with a common tag on the **Explore** page.
6. Select the **Configuration** tab.

   The design workspace for the dictionary opens. The workspace is a grid of rows and columns.
7. Enter the dictionary data.

   Begin in the first row of column 1. The dictionary can have different numbers of values on each row.

   Column 1 is the default valid value column. Populate column 1 with the values that a transformation will return by default at run time.

   Note: You can design a dictionary in which more than one column might serve as a valid column. When you add a dictionary to an asset, you can select any column as the valid column.
8. Save the dictionary.

You can also populate the dictionary by "Importing dictionary data from a file" on page 11.

Adding data to a dictionary

You can create dictionaries with any data that is useful to the data project that you work on.

When you create a dictionary, the **Configuration** view displays fifty empty rows by default. You can enter data into any cell in any row. The dictionary does not evaluate the data that you enter, and the values in each column do not need to be objectively correct or incorrect. As best practice, enter data to the first cell in the first available row. After you add the data, save the dictionary.

You can add data to a dictionary in the following ways:

- Select an empty cell in the dictionary, and type data into the cell.

  If the dictionary does not contain an empty row, use the **Add Row** option to add rows to the dictionary. The rows appear at the bottom of the dictionary.
- Copy and paste data from a list or table in another application, such as a web page.

  You can paste the data to empty rows in the dictionary, or you can overwrite the current dictionary data.

  You can also copy and paste data within the dictionary and paste data from another dictionary.

  You do not need to add rows to accommodate data that you paste into a dictionary. The dictionary expands to accommodate the data that you add.
- Import data from a delimited file or a Microsoft Excel file.

  For more information on importing data, see "Importing dictionary data from a file" on page 11.

Updating data values

To update the value in a cell, select the cell and type a new value. You can select cells or rows in a sequence and delete the data that the cells or rows contain.

Use Windows shortcuts to copy, paste, and delete data.

If you create a dictionary in Reference 360, you can view the contents of the dictionary in Data Quality or Data Integration, but to edit the dictionary data, you must open the dictionary source in Reference 360.

Importing dictionary data from a file

You can import dictionary values from a Microsoft Excel file or any file that uses a delimiter that Data Quality or Data Integration recognize. You can import the file data to an empty dictionary or to a dictionary that contains data. When you import the file data, Data Quality or Data Integration add the data to the first empty row in the dictionary.

If you import data from a file with multiple columns, verify that the valid data column in the file matches the valid data column in the dictionary. The first column is the valid value column by default.

1. From the **Explore** page, select and open a dictionary.

   Or, click **New** and create a dictionary.

   If you create a dictionary, complete the fields on the Definition tab.
2. Select the Configuration tab.
3. Select the option to import data from a file.
4. In the **Import a flat file** dialog box, complete the following steps:
   - Choose the file that contains the data to import.
   - Select or clear the option to import column names. The option is cleared by default.
   - Select the line in the file from which to begin the data import. By default, data is imported from the first line in the file.
   - If you import data from a delimited file, select the delimiter. By default, a comma delimiter is used.
   - If the file uses single quotes or double quotes as a text qualifier, select the text qualifier. By default, text qualifiers aren't recognized.

   Note: A dictionary can contain a maximum of 42 columns. If you try to import a file with additional columns, you are prompted to reduce the number of columns in the file.
5. Click **Import**.
6. Save the dictionary.

   Note: When you import a very large dictionary, the dictionary may be saved after the import operation is complete. A very large dictionary may contain up to 55 MB of data.

Rules and guidelines for importing data

Consider the following rules and guidelines when you import data:

- The import option supports CSV and Microsoft Excel files. The file must use the UTF-8 character encoding.
- You can import up to 55 MB of data from a file.
  To add very large quantities of data to a dictionary, use the file import mechanism. Do not cut and paste very large quantities of data into a dictionary.
- You can preview up to 100 lines of data before you import.
- If a dictionary contains more than 100,000 rows, pagination applies to the asset and the rows are displayed on multiple pages. Each page displays 100,000 rows.

- When the quantity of data that you import exceeds the pagination threshold for dictionaries, the dictionary is automatically saved. For information about pagination in dictionaries, see "Working with very large dictionaries" on page 14.
- A dictionary asset page can paginate and display a maximum of approximately 500,000 rows of dictionary data. The physical limit for the dictionary depends on the quantity of data that the dictionary contains. You can use the query options to find and work with any data that the dictionary stores beyond the pagination limit. The complete dictionary data set remains available to assets that you test and to transformations that contain data quality assets.

  The dictionary configuration pane displays the number of rows in the dictionary.

Dictionary conversion options

When you create a dictionary through the Data Quality or Data Integration user interface or through the Data Quality Dictionary API, you can update the dictionary from either the user interface or through the API.

A dictionary cannot accept updates through the user interface and the API simultaneously. You can convert the dictionary from one editing environment to another. Enforcing a single environment for dictionary updates ensures that users in different environments do not inadvertently overwrite each other's changes.

When you open a dictionary, the **Definition** and **Configuration** tabs display an option to switch to the alternative environment. To change the editing environment from the Data Quality Dictionary API to the user interface, click **Convert to UI**. To change the editing environment from the user interface to the Dictionary API, click **Convert to API**.

After you convert the dictionary, a message appears to indicate that the dictionary will be saved and converted to the environment you specified.

For more information about how to update a dictionary through the Data Quality Dictionary API, see the *API Reference* help in Data Quality.

Downloading dictionary data

Download the data in a dictionary to a comma-separated file.

1. From the **Explore** page, select and open a dictionary.
2. Select the **Configuration** tab.
3. Click the **Download** icon.

   By default, the name of the downloaded file is same as the dictionary name.

Note the following points when you download the dictionary data:

- The operation downloads the complete contents of a dictionary and disregards any filter that you apply to the dictionary before you download.
- The data might contain characters that spreadsheet applications can interpret as formulas.
- You can't download a dictionary that you created in Reference 360.

Finding and filtering data values

Use the find and replace options to update one or more values in a dictionary. Filter the dictionary rows to review the data values that match your criteria and to ensure that you update only the values that you require.

Note: The find, replace, and filter options operate on the page that currently displays the dictionary data. Additionally, you can filter the complete contents of a very large dictionary. The **Add Filter** option appears when you can filter data across the complete dictionary.

Finding and replacing data values

You can find and replace data values throughout a dictionary, or you can find and replace data values in one or more rows or columns that you select.

1. To display the find and replace options, click **Find**.
2. Optionally, select one or more rows or columns to restrict the find and replace operations.

   To select a row, click the row ID. To select a column, click the column name. Use the Control or Shift keys to select multiple rows or columns.
3. Enter a value in the Find field and click the search arrow.

   The matching dictionary values are highlighted in yellow. Additionally, the cell that contains the first instance of the value is selected.
4. Perform one of the following actions:
   - To replace a single value, type in the cell that contains the value. Or, enter the new value in the Replace field and click **Replace**.

     When you click **Replace**, the current cell is updated and the next cell in the dictionary that contains the value is selected.
   - To replace successive instances of the value, click **Replace** each time a cell is selected.

     To skip a cell, or to navigate through the search results, use the Up and Down arrows beside the Find field.
   - To replace all instances of the value in the dictionary, click **Replace All**.

     Before you click **Replace All**, review the dictionary to ensure that you want to replace every instance of the value. The Find field displays the number of times that the value occurs in the dictionary.
5. After you replace the data values, save the dictionary.

Note: The find and replace operations replace only the exact set of characters you enter. If the characters appear within a longer string in a cell, only the matching characters are replaced, while the rest of the string remains unchanged.

If the characters appear more than once in a cell, all instances of the characters in the cell are replaced.

In cases where the characters appear more than once in a cell, each occurrence is included in the results of the Find field.

Filtering data values

When you apply a filter to a dictionary, only the rows that match the filter you set are displayed, and all other rows are hidden.

You can set a filter on each column in a dictionary. When you set multiple filters, the rows that match all of the filters are displayed.

The filter options are hidden by default. To display the options, click **Filter**. When you click **Filter**, the **Add Field** options appear.

1. Click **Add Field** and select a column from the column list.

2. Enter a filter value in the **Value** field, and click **OK**.

   The dictionary displays the rows that contain the filter value in the column that you specify.

   To add another filter, repeat the steps.

When you add one or more filters to a dictionary, the find and replace operations apply only to the data rows that meet the filter criteria.

Note: You can also select one or more cells, rows, or columns when you apply a filter. The find and replace operations are performed within the cells, rows, or columns that you select.

Working with very large dictionaries

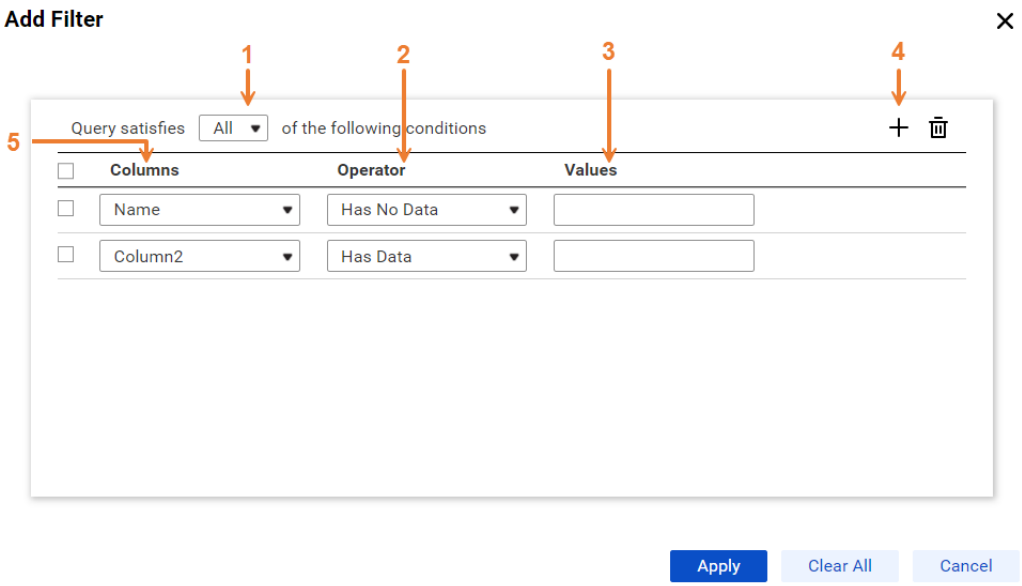
Up to 500,000 rows of dictionary data can be paginated at a time, based on five columns of dictionary data. The physical limit for your dictionary depends on the number of rows and columns in the dictionary and the number of characters in each cell. The limit governs any operation to view or update data, such as a search and replace operation, that you perform in the dictionary.

When a dictionary contains more data than what can be paginated in a single operation, you can use the filter query options to retrieve a subset of the dictionary data. The filter is applied to the complete dictionary, regardless of the quantity of data that the dictionary may contain. Use the **Add Filter** option to define the filter query.

A single filter query can return data up to the pagination limit. You can define multiple filters if necessary to cover the complete contents of your dictionary.

Note: The complete dictionary data set is available to assets that you test and to transformations that contain data quality assets.

The following image shows the options that you use to create a filter query:



The dialog box contains the following options and attributes:

1. Query scope option.
   Determines whether the dictionary data that the query retrieves must meet any condition that you add or all conditions that you add.
2. Operator attribute.

   Indicates the operator for each condition. The operator determines how the query analyzes the column data. You can select the following operations:

   - Equals
   - Not Equals
   - Contains
   - Not Contains
   - Has Data
   - Has No Data
3. Values attribute.

   Specifies the value that the condition looks for in a column when you select one of the following operators: Equals, Not Equals, Contains, or Not Contains.
4. Add option.

   Adds a condition to the query. You can add one or more conditions.
5. Columns attribute.

   Indicates the column to which each condition applies.

Running a query on a large dictionary

Create a filter query to retrieve data from the full set of dictionary values. The filter options are available in very large dictionaries.

For information about very large dictionaries and the pagination limits that can apply to them, see "Working with very large dictionaries" on page 14.

You build a filter query on dictionary column values. For example, you can create a query that can retrieve rows with given values in a column that you select.

1. On the **Configuration** view, click **Add Filter**.
2. In the **Add Filter** dialog box, click the Add icon to add a condition.
3. Choose a column and an operator for the condition. If required, add a value that the query can look for in the column.
4. Add more conditions if required.
5. If you add multiple conditions, choose one of the following options:
   - All. Any row that meets all of the conditions is retrieved.
   - Any. Any row that meets at least one of the conditions is retrieved.
6. Click **Apply**.

Note: To edit a query, click **Edit Filter** in the **Configuration** pane.

Verifying the dictionary download location

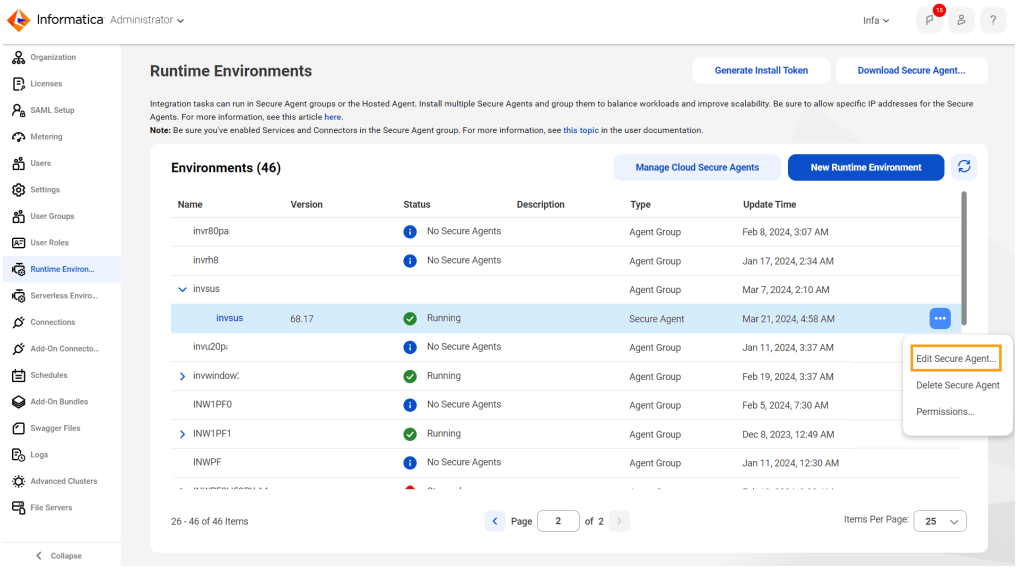
The Secure Agent can download a dictionary when you test an asset that specifies a dictionary or when you run a mapping or profile that reads a dictionary. You can verify or update the download location in the Administrator service.

If you plan to update the download location, confirm the update with an Admin user for your organization.

1. From the **My Services** page, select the **Administrator** service.
2. Choose the **Runtime Environments** option.
3. Select the **Secure Agent** that will test or run the assets, profiles, or mappings.

   Note: Each Secure Agent has a separate set of properties.
4. Hover over the **Actions** icon for the Secure Agent, and select the **Edit Secure Agent** option.

   The following image shows the option:



   The **Secure Agent** page appears.
5. Under **System Configuration Details**, select the following options:
   - Select **Data Integration Server** in the **Service** field.
   - Select **IDQ** in the **Type** field.

   The **System Configuration Details** pane returns a list of properties based on the type that you specified.
6. Find the *ReferenceDataLocation* property in the **Name** column.

   The corresponding entry in the **Value** column shows the current download directory.

For more information about dictionary downloads, see "Dictionary operations" on page 7.

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